

A Macroscopic-Mass Test of Density-Gated Scalar Fields in a Metre-Scale Open Vacuum

Working draft for expert critique. Not peer reviewed. Gravity Innovation (independent project) · v2, September 2026 · Contact: Paul Wilkerson (please open an issue on this repository) · Code and outputs: <https://github.com/paulwilkerson1985-cmd/gravity-innovation>

Disclosure: produced by a non-academic project with extensive AI assistance (Anthropic's Claude, including independent derivations and adversarial internal reviews). Every number here needs expert verification.

Summary

Scalar fields whose vacuum value depends on the size of the empty region around them (symmetrons and their generalizations, environment-dependent dilatons, chameleons) have only been tested in chambers of radius ≤ 6 cm, or with sources outside the vacuum or behind shields. Any such field with a vacuum range above ~ 2 cm never switches on in existing apparatus.

- **Proposal.** A torsion balance whose source and test bodies share a ≥ 1 m **unshielded vacuum**. For the benchmark symmetron ($1/\mu = 2\text{--}30$ cm) the force between screened bodies is **independent of the coupling scale M** , and environmental switch tests give a fingerprint no artifact shares. It is the macroscopic-mass extension, one decade up in range, of Clements et al. (2024; Burrage's group), who proposed condensing the symmetron in a 10 cm UHV chamber and detecting its domain walls with atoms or nanobeads at $1/\mu = 1\text{--}10$ mm.
- **Bounds on the benchmark.** SN1987A (our re-derivation): **$M \gtrsim 6$ TeV (5-8 across core profiles)** for the universal coupling, ≈ 4.5 TeV (3.8-5.8) for a nucleon-mass-only coupling. A Boltzmann calculation of the thermal relic, against the 2026 BBN+CMB+BAO limit, excludes $M \leq 7.5$ TeV (5.9-9.6) for the universal coupling if reheating exceeded a few GeV. Air pinning gives the ceiling $M \leq 3.65 \text{ TeV} \times (1/\mu / \text{cm})$. Surviving wedge: **$1/\mu \approx 2\text{--}30$ cm, $M \gtrsim 7.5$ TeV ($\gtrsim 6$ TeV for any thermal history) up to the ceiling** (Fig. 1A).
- **Publishable either way.** A Tier-1 null gives $v^2 < 2\text{--}3 \times 10^{-11}$ N at $1/\mu = 4\text{--}10$ cm, the first macroscopic-mass bound in the band and about three decades below the only partial constraint (HUST-09); Tier 2 reaches five. Tier 2 also excludes a plausibly open (unverified) corner of the environment-dependent dilaton and, with a heavy attractor, improves inverse-square-law limits at 10-30 cm by $\sim 10\text{--}100\times$.

1. The class, the benchmark, and why the regime is untested

Benchmark. $V(\phi) = -\mu^2\phi^2/2 + \lambda\phi^4/4$, universal coupling $A(\phi) = 1 + \phi^2/2M^2$, vacuum value $v^2 = \mu^2/\lambda$. Between screened bodies $F = v^2 \tilde{F}(\text{geometry})$, $1 \text{ eV}^2 \triangleq 8.12 \times 10^{-13} \text{ N}$.

Where the field is pinned ($\phi \approx 0$). Inside dense matter; in any vacuum below a size threshold (empty cylinder $(2.405/R)^2 + (\pi/L)^2 < \mu^2$; with the test bodies our validated solver gives $R = L/2 \approx 3.2\text{--}3.7/\mu$, hardware adds $\sim 12\%$); in room air if $\mu M < 7.2 \times 10^7 \text{ eV}^2$; behind foils once $t \gg t^* = \mu M^2/\rho \approx 1.3 \mu\text{m} \times (1/\mu / \text{cm})$ for Cu at the air ceiling. Anything solid thicker than ~ 0.1 mm pins the field; only μm foils and $\leq 20 \mu\text{m}$ wires are transparent.

The class. The threshold belongs to every Z_2 -breaking scalar with a quadratic matter coupling (generalized symmetrons, the asymmetron, the dark-matter variant of Burrage et al. 2019). The environment-dependent dilaton (Brax, Fischer, Kading & Pitschmann 2022) has no vacuum threshold, but its force grows with chamber size and vanishes in a 12 cm chamber (§3). The $n = 1$ dark-energy chameleon gives an M -independent screened-sphere force of 2.4×10^{-13} N in a 1 m chamber at Λ_{DE} , just below our Tier 2 (3×10^{-13} N), but that region is already excluded for all M (Jaffe et al. 2017; Yin et al. 2022): a cross-check, not new space.

Desk study (~ 40 experiments, incl. Heyl 1930 and Overstreet et al. 2022). Atom interferometers: sources outside the vacuum or chambers of radius ≤ 6 cm. Torsion balances and G measurements: sources outside the vacuum or behind shields; the partial exception, HUST-09 (open-top shield; Tu et al. 2010), excludes only $v^2 \gtrsim 1\text{--}5 \times 10^{-8}$ N at $1/\mu \approx 1.5\text{--}4$ cm by our 3D model ($\times 3$). Casimir, levitated sensors,

neutrons, spectroscopy: $\mu \gtrsim 0.1$ meV or mm-scale cages. The loophole is known (Upadhye 2013; Burrage et al. 2016; Chiow & Yu 2020; Banks, Carlton, Elder, Hird & McCabe 2026 assume a 5 cm vacuum radius); we found no proposal exploiting it with macroscopic masses.

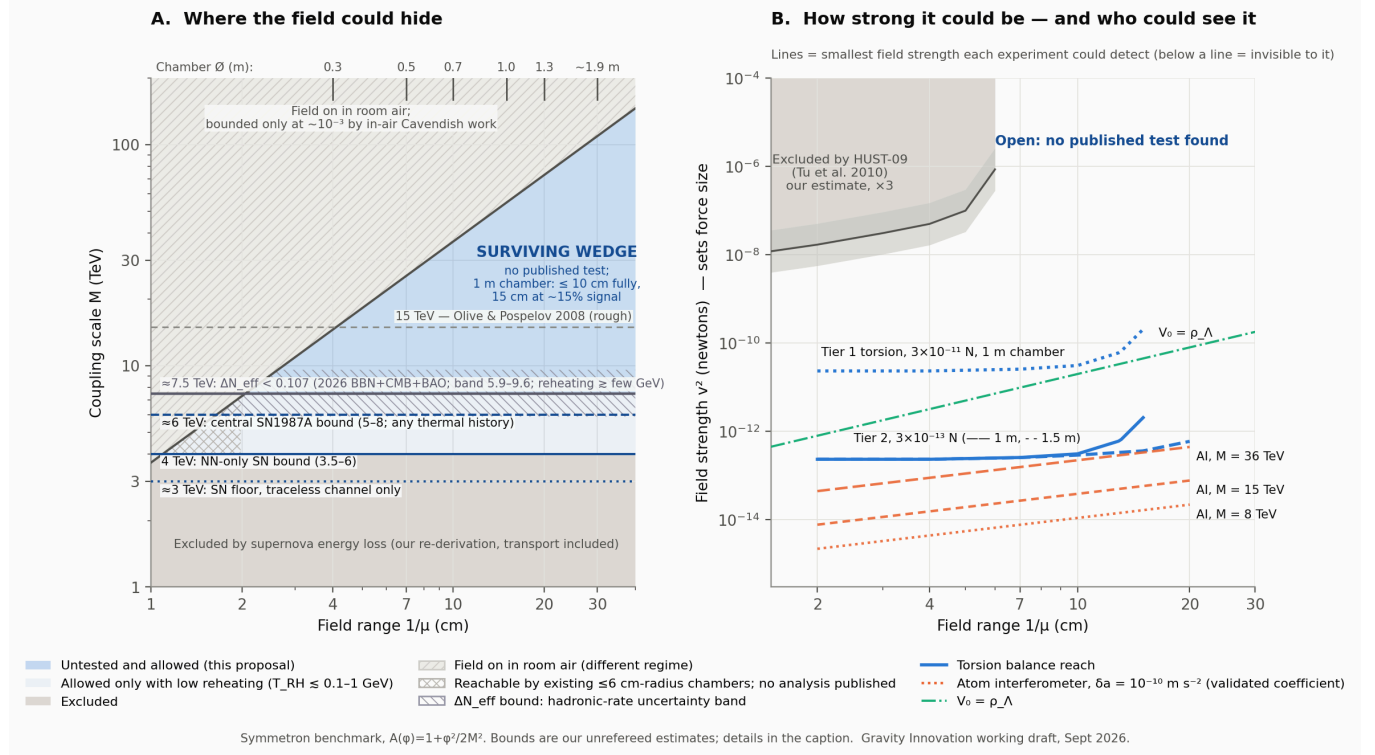


Figure 1. (A) Range versus coupling scale M . Gray: excluded by SN1987A (ours, unrefereed; lines at the central 6 TeV, the NN-only 4 TeV and the traceless-channel 3 TeV). Solid line and hatched band: the ΔN_{eff} bound, 7.5 TeV (5.9–9.6 across the hadronic-rate band) if reheating exceeded a few GeV; the pale region below it survives only with low reheating (and, below 6 TeV, only with the NN-only SN bound). For the nucleon-mass-only operator the SN lines scale by 0.72 and the ΔN_{eff} bound does not apply. Above the diagonal the field is on in room air ($\mu M > 7.2 \times 10^7$ eV 2 ; bounded only at $\sim 10^{-3}$ by in-air Cavendish work). Cross-hatch: reachable by existing ≤ 6 cm-radius chambers, no analysis published. Top ticks: minimum chamber diameter with the test bodies (hardware +12%). (B) v^2 versus range: HUST-09 exclusion (ours, $\times 3$); the dark-energy line $V_0 = \rho \Lambda$; torsion-balance reach (Tier 1: 3×10^{-11} N; Tier 2: 3×10^{-13} N, solid 1 m and dashed 1.5 m chamber; $S/S_0 = 0.81$, membrane loss not included); atom-interferometer reach at $\delta a = 10^{-10}$ m s^{-2} for $M = 8, 15, 36$ TeV.

2. Bounds on the benchmark

SN1987A (our re-derivation; companion note). We bound the universal coupling $\varphi^2 T_{\mu}^{\mu}/2M^2$. Olive & Pospelov (2008) obtained $M \gtrsim 15$ TeV by assuming an $O(1)$ change of scalar charge per NN collision. Energy-momentum conservation removes monopole and dipole pair emission, suppressing the NN rate by $\sim 9(T/m_N)^2 \approx 10^{-2}$. The dominant trace channel is fixed at $\omega \rightarrow 0$ by the Wigner time delay (verified to 0.5% at 10 MeV and 0.1% above 75 MeV with the AV18, Reid93 and Nijmegen-II potentials); with finite- ω matrix elements the trace/traceless ratio is $K_T = 12$ (10–13) (Maxwell-Boltzmann), 6.7–8.1 with Pauli blocking. $\pi\pi \rightarrow n\phi$ is not suppressed and dominates for $Y_{\pi} \gtrsim 0.2\%$. Transport is treated by Monte Carlo.

Assumption	Bound on M (transport included)
Traceless channel only (floor)	≈ 3 TeV (2.2–3.9)
NN only	3.5 (cold profile) – 6.0 (hot) TeV
Central: NN + pions, $Y_{\pi} \approx 1\%$	5.3–7.9 TeV $\rightarrow$ "M $\gtrsim 6$ TeV (5–8)"; we would defend 5–6 TeV in print
Coupling to nucleon mass only (Olive–Pospelov's operator)	$\approx 0.72 \times$ the above: ≈ 4.5 TeV (3.8–5.8)

The excluded band extends down to 1–2 TeV: a lower bound, not an island. *Credit*: the conservation-law suppression was noted in 2007 for $d_U = 2$ unparticles (Hannestad, Raffelt & Wong; Dutta & Goyal; Freitas & Wyler); the traceless channel is the dilaton low-energy theorem of Hanhart, Phillips, Reddy & Savage (2001). New here: the exact pair identity with its κ -weights, the time-delay trace channel, the pion channel and the transport. Dominant uncertainties: Y_π and in-medium pion physics ($\times 0.75$ –1.3 in M ; Fore et al. 2024 find the π^- mass *rises* in neutron-rich matter) and the parametrized profiles ($\times 0.8$ –1.2).

Thermal relic (Boltzmann calculation). In the early-universe symmetric phase the field is light and stays in contact with the plasma through the T^μ_μ coupling: the gluon trace anomaly above the QCD crossover, $\pi\pi \rightarrow \varphi\varphi$ below it (vertex $\langle 0|\Theta|\pi\pi\rangle = s + 2m_\pi^2$, Omnès-unitarized). Solving the Boltzmann equation with a lattice equation of state gives $\Delta N_{\text{eff}} = 0.24$ (0.14–0.32) at $M = 5$ TeV, 0.17 (0.10–0.27) at 6 TeV and 0.07 at 10 TeV; φ stays partly coupled through the crossover, so our earlier instantaneous-decoupling estimate was too low above 5 TeV. The 2026 BBN+CMB+BAO limit $\Delta N_{\text{eff}} < 0.107$ (95%; Goldstein & Hill 2026) then excludes **$M \lesssim 7.5$ TeV (5.9–9.6 across the hadronic-rate band)** if reheating exceeded a few GeV (6.2–6.6 TeV for $T_{\text{RH}} \approx 0.2$ –0.5 GeV). It drops below the SN bound for $T_{\text{RH}} \lesssim 100$ MeV and does not apply to the nucleon-mass-only operator (no pion channel; $\Delta N_{\text{eff}} \approx 0.06$). The dominant uncertainty is the $\pi\pi$ trace form factor at $\sqrt{s} \approx 0.5$ –1 GeV; CMB-S4 ($\sigma \approx 0.03$) would reach $\Delta N_{\text{eff}} \approx 0.06$, i.e. $M \approx 10$ TeV. The Z_2 transition itself ($T \sim 100$ –200 keV) releases $\sim 10^{-13}$ eV⁴ and is inert. Today $\rho_{\text{crit}} = \mu^2 M^2 \approx 5 \times 10^{-3}$ –1 kg m⁻³, so the field is on above ~ 0 –40 km and in space; all bodies being screened and the range ≤ 30 cm, we found no lunar-laser-ranging or satellite bound.

Naturalness. $\delta\mu^2 \sim \Lambda_{\text{UV}}^4/16\pi^2 M^2$: for $M = 5$ –10 TeV, $\mu = 10^{-5}$ eV the required cancellation is $\sim 10^{-19}$ (1 GeV cutoff), 10^{-28} (200 GeV), 10^{-31} (1 TeV). The transparent completion, a Higgs portal with $\lambda_p = m_h^2/2M^2 \sim 10^{-4}$, induces $m_\varphi \sim \text{GeV}$. This is the tuning of every laboratory symmetron and chameleon (Upadhye, Hu & Khoury 2012); the mitigation routes in print (Burrage, Copeland & Millington 2016; Burrage et al. 2018) do not reach $\mu\text{eV/TeV}$. We treat the model as a phenomenological parametrization of a documented blind spot, and the tuning as the honest prior.

3. Proposed experiment

Benchmark signal (validated axisymmetric solver). Cu spheres of 1 kg and 8.1 kg, centres 14 cm apart, 1 m chamber, $1/\mu = 10$ cm: $\vec{F} = 1.195 \pm 0.005$, i.e. $\delta G/G = 4.3 \times 10^7 \times v^2[\text{N}]$. A 3D finite-volume calculation of the dumbbell torque gives 0.90 ± 0.04 of the pairwise proxy; hardware (turntable $\geq 1.5/\mu$ from the bodies) costs a further 9–13%; reach lines use $S/S_0 = 0.81$ (membrane loss not included).

Design principle: the screened-body force is set by surface geometry, not mass. It saturates once the wall exceeds $\sim 10 t^*$, while gravity grows with mass. So: **hollow Cu shells** ($r = 3$ cm, 2 mm wall, 0.2 kg) as test bodies and **Cu foil disks** ($r = 5$ cm; 23–68 μm , 1.6–4.8 g at $M = 15$ TeV) 4 cm away as the attractor. The foil pulls the shell with 1.04 – $1.29 v^2$ (the 8.1 kg sphere at 14 cm gives $1.20 v^2$) with a Newtonian pull of 3 – 10×10^{-12} N: **13–30 $\times$ the Tier-2 signal instead of $\sim 10^3 \times$** . A foil-thickness series separates the two (F_{scalar} saturates, $F_N \propto t$) and measures M ; the high- M edge needs 130 μm at 36 TeV.

Tiers. Tier 1: 3×10^{-11} N (borrowed ≥ 1 m chamber; $\approx$ \$100–130k, one student-year). Tier 2: 3×10^{-13} N (established torsion-balance group; thermal-noise SNR ≈ 40 per hour).

Electrostatic baseline. A permanent, grounded, **full-width 2 μm Al-Mylar membrane** between attractor and pendulum screens DC fields at 95–99%. As a Robin sheet it has $\kappa/\mu \approx 0.14 \times (1/\mu / 10 \text{ cm})(15 \text{ TeV}/M)^2 \approx 11\%$ signal loss at the benchmark (10 cm, 15 TeV) and $\leq 12\%$ wherever $M \geq 5 \text{ TeV} \times \sqrt{(1/\mu / \text{cm})}$, but 17% (4 cm), 37% (10 cm) and 75% (30 cm) along the 7.5 TeV floor, where a thinner film (0.9 μm Mylar + 30 nm Au, $\times 0.63$ in κ) is needed. It must span the chamber (a framed $1.5/\mu$ membrane halves the signal). Bias nulling and Au surfaces are the check; Tier 2 needs $\Delta V \lesssim 30$ mV.

Switch: liner or end plates. An open tube of radius $2.8/\mu$ around the pendulum axis, or end plates moved from the chamber ends to $\pm 1.5/\mu$, drives the field to zero with every mass fixed, Newtonian-neutral by symmetry, no gas. Run liner-in/out with the attractor running: the attractor-synchronous torque is Newtonian + scalar (out) versus Newtonian only (in), so the Newtonian part cancels to the geometric reproducibility. A one-sided movable foil is *not* a switch: it pulls the test body with 1.2 – $1.5 v^2$ of its own.

Fingerprint. A genuine signal must pass all of: (1) liner switch: zero below threshold; (2) foil-thickness saturation curve (M); (3) gas shape test, Tier 1 only: $S/S_0 \approx (1 - \rho/0.584 \rho_{\text{crit}})^{1.15}$, vanishing at $\rho = 0.584 \mu^2 \text{M}^2$ (second M handle), identical for He and Xe at matched *mass* density; a lateral 1 mK gradient across 1 m drives convection torques $\sim 70\times$ Tier 1, so this is a shape test, not a modulation; (4) composition independence (Al vs W) to $O(\mu/m_{\text{in}}) \approx 1\text{--}2\%$; (5) distance law: $10.9\times$ fall from 14 to 30 cm in the benchmark geometry versus $4.6\times$ Newtonian; (6) domain walls: none metastable at chamber radius $5\text{--}7/\mu$ (our choice); at $\geq 8/\mu$ a pinned wall changes the force by 3–5% and gives a bimodal run-to-run distribution. Systematics: patch potentials (large-scale Volta variation of the rotating source, insulator charging), foil-shell distance drift, gas-run radiometric forces, tilt, seismic noise.

Reach and by-products. A Tier-1 null: $v^2 < 2.3 / 2.6 / 3.1 \times 10^{-11} \text{ N}$ at $1/\mu = 4 / 7 / 10 \text{ cm}$, reaching the dark-energy line $V_0 = \rho_\Lambda$ ($v^2 = 2 \times 10^{-11} \text{ N} \times (1/\mu / 10 \text{ cm})^2$) near $1/\mu \approx 10 \text{ cm}$; Tier 2 is $100\times$ lower.

Dilaton: solving the Liouville equation for $V = V_0 e^{-\{\lambda\phi/M_{\text{Pl}}\}}$, $A = 1 + A_2\phi^2/2M_{\text{Pl}}^2$ on the same grid with $V_0 = \rho_\Lambda$, the benchmark force peaks at $1.2 \times 10^{-11} \text{ N}$ at $\lambda = 10^{27}$ ($3 \times 10^{-11} \text{ N}$ in a 2 m chamber), and Tier 2 excludes $5 \times 10^{25} \leq \lambda \leq 2.5 \times 10^{28}$ for $A_2 \gtrsim 10^{28}(\lambda/10^{27})^2$ ($M_{\text{eff}} \lesssim 25 \text{ TeV}$). Published dilaton plots resolve $A_2 \sim 10^{40}$ and the cosmological subspace, not this corner: plausibly open, unverified. **Inverse-square law:** with the 8 kg attractor, Tier 2 corresponds to $|\alpha| \approx 1.8 \times 10^{-5}$ ($\lambda = 10 \text{ cm}$) and 1.2×10^{-5} (30 cm) against current $\sim 10^{-4}\text{--}10^{-3}$ (Hoskins et al. 1985; Adelberger, Heckel & Nelson 2003); Tier 1 is comparable to existing limits. **Atom interferometry (coefficient validated):** unscreened atoms feel $a = C \cdot 1.8 \times 10^{-8} \text{ m s}^{-2} \cdot (v^2/\text{eV}^2)(10 \text{ cm}/(1/\mu))(5 \text{ TeV}/\text{M})^2$, $C = 1.06$ at 2–3 cm above a 6 cm sphere in a 1 m chamber, giving $v_{\text{min}}^2 = 4.3 \times 10^{-15} \text{ N} \times (1/\mu / 10 \text{ cm})(\text{M}/5 \text{ TeV})^2$ at $\delta a = 10^{-10} \text{ m s}^{-2}$. The interferometer covers the low-M part of the wedge ($M \lesssim 40 \text{ TeV}$ at 10 cm), the M-independent balance the high-M part; they want different attractors (compact sphere versus foil).

4. Caveats

Nothing here is refereed: the SN and ΔN_{eff} analyses are ours (the latter assumes a standard thermal history), and the dilaton corner is unverified. The prior probability of a detection is well under 1%, set by the naturalness numbers; the value is a publishable exclusion of a documented gap, the two by-products, and the technique. A positive result would be a new force with metrological uses (a calculable, switchable reference force: $\leq 10^{-5} \text{ N}$, attractive, two-body and vacuum-only).

5. What we ask

A 30-minute call or an email on three points: (1) **Solver check:** run SELCIE (or your own code) on the benchmark (Cu spheres 1 kg and 8.1 kg, centres 14 cm apart, 1 m $\times$ 1 m cylinder, $1/\mu = 10 \text{ cm}$; we get $\tilde{F} = 1.195 \pm 0.005$ for the force on the 1 kg test sphere) and on the membrane case (a full-width Robin sheet with $\kappa/\mu = 0.1$ midway between the sphere surfaces: 7.8% loss; real 2 μm Al-Mylar at 15 TeV is $\kappa/\mu \approx 0.14$, $\approx 11\%$); our repository reproduces both in under a minute. (2) **Missed experiments:** any experiment or proposal with two screened bodies sharing an unshielded vacuum larger than $\sim 7/\mu$ at $1/\mu \geq 2 \text{ cm}$? (3) **The dilaton corner:** is ($\lambda \sim 10^{26}\text{--}10^{28}$, $A_2 \sim 10^{28}\text{--}10^{31}$) already excluded?

Key references

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